

A second consideration for transitioning pilots concerns turbine engine heat sensitivity. A turbine engine cannot tolerate an over temperature condition for more than a very seconds without experiencing serious damage. Engine temperatures get hotter during starting than at any other time. Thus, turbine engines have minimum rotational speeds for introducing fuel into the combustion chambers during startup. Vigilant monitoring of temperature and acceleration on the part of the pilot remain crucial until the engine is running at a stable speed. Successful engine starting depends on assuring the correct minimum battery voltage before initiating start or employing a ground power unit (GPU) of adequate output.

After fuel is introduced to the combustion chamber during the start sequence, “light-off” and its associated heat rise occur very quickly. Engine temperatures may approach the maximum in a matter of 2 or 3 seconds before the engine stabilizes and temperatures fall into the normal operating range. During this time, the pilot should watch for any tendency of the temperatures to exceed limitations and be prepared to cut off fuel to the engine.

An engine tendency to exceed maximum starting temperature limits is termed a hot start. The temperature rise may be preceded by unusually high initial fuel flow, which may be the first indication the pilot has that the engine start is not proceeding normally. Serious engine damage occurs if the hot start is allowed to continue.

A condition where the engine is accelerating more slowly than normal is termed a hung start or false start. During a hung start/false start, the engine may stabilize at an engine rpm that is not high enough for the engine to continue to run without help from the starter. This is usually the result of low battery power or the starter not turning the engine fast enough for it to start properly.

Takeoffs in turboprop airplanes are not made by automatically pushing the power lever full forward to the stops. As stated earlier, depending on conditions, takeoff power may be limited by either torque or by engine temperature. Normally, the power lever position on takeoff is somewhat aft of full forward.

Takeoff and departure in a turboprop airplane (especially a twin-engine cabin-class airplane) should be accomplished in accordance with a standard takeoff and departure “profile” developed for the particular make and model. [Figure 15-11] The takeoff and departure profile should be in accordance with the airplane manufacturer’s recommended procedures as outlined in the Federal Aviation Administration (FAA)-approved Airplane Flight Manual and/or the Pilot’s Operating Handbook (AFM/POH). The increased complexity of turboprop airplanes makes the standardization of procedures a necessity for safe and efficient operation. The transitioning pilot should review the profile procedures before each takeoff to form a mental picture of the takeoff and departure process.

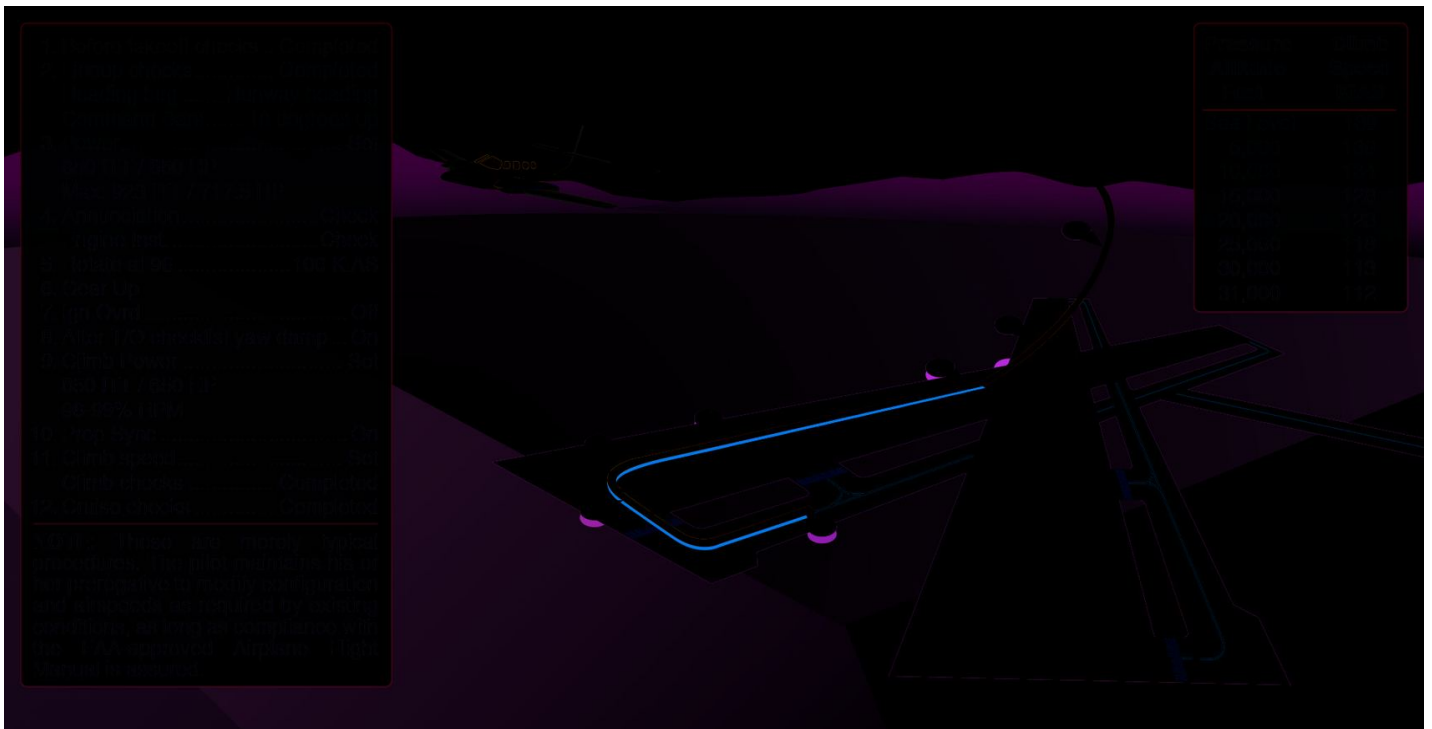


Figure 15-11. Example of a typical turboprop airplane takeoff and departure profile.

For any given high-horsepower operation, the pilot can expect that the engine temperature will climb as altitude increases at a constant power. On a warm or hot day, maximum temperature limits may be reached at a rather low altitude, making it impossible to maintain high horsepower to higher altitudes. Also, the engine's compressor section has to work harder with decreased air density. Power capability is reduced by high-density altitude and power use may have to be modulated to keep engine temperature within limits.

In a turboprop airplane, the pilot can close the throttles(s) at any time without concern for cooling the engine too rapidly. Consequently, rapid descents with the propellers in low pitch can be dramatically steep. Like takeoffs and departures, approach and landing should be accomplished in accordance with a standard approach and landing profile. [Figure 15-12] However, when flying an airplane equipped with a split shaft/free turbine engine, the pilot should anticipate the demand for power and account for any lag in “spool-up” time.

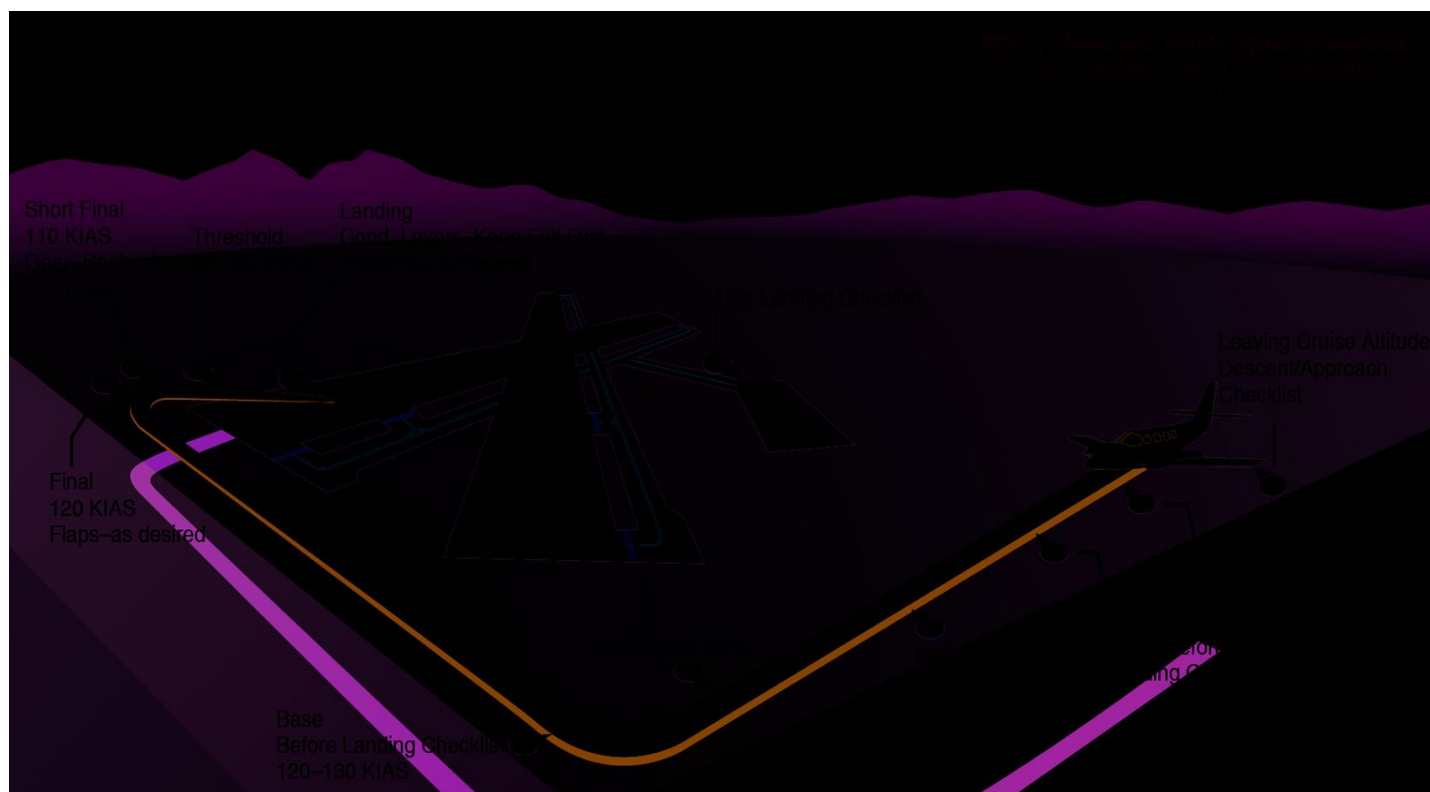


Figure 15-12. Example of a typical turboprop airplane arrival and landing profile.

A stabilized approach is an essential part of the approach and landing process. In a stabilized approach, the airplane, depending on design and type, is placed in a stabilized descent on a glidepath ranging from 2.5 to 3.5°. The speed is stabilized at some reference from the AFM/POH—usually 1.25 to 1.30 times the stall speed in approach configuration. The descent rate is stabilized from 500 fpm to 700 fpm until the landing flare.

Landing some turboprop airplanes (as well as some piston twins) can result in a hard, premature touchdown if the engines are idled too soon. This is because large propellers spinning rapidly in low pitch create considerable drag. In such airplanes, it may be preferable to maintain power throughout the landing flare and touchdown. Once firmly on the ground, propeller beta range operation dramatically reduces the need for braking in comparison to piston airplanes of similar weight.

Training Considerations

The medium and high altitudes at which turboprop airplanes are flown provide an entirely different environment in terms of regulatory requirements, airspace structure, physiological requirements, and even meteorology. The pilot transitioning to turboprop airplanes, particularly those who are not familiar with operations in the high/medium altitude environment, should approach turboprop transition training with this in mind. Thorough ground training should cover all aspects of high/medium altitude flight, including the flight environment, weather, flight planning and navigation, physiological aspects of high-altitude flight, oxygen and pressurization system operation, and high-altitude emergencies.

Flight training should prepare the pilot to demonstrate a comprehensive knowledge of airplane performance, systems, emergency procedures, and operating limitations, along with a high degree of proficiency in performing all flight maneuvers and in-flight emergency procedures. The training outline below covers information used by pilots to operate safely at high altitudes.

Ground Training

1. High-Altitude Flight Environment
 - a. Airspace and Reduced Vertical Separation Minimum (RVSM) Operations
 - b. Title 14 Code of Federal Regulations (14 CFR) part 91, section 91.211, Requirements for Use of Supplemental Oxygen
2. Weather
 - a. Atmosphere
 - b. Winds and clear air turbulence
 - c. Icing
3. Flight Planning and Navigation
 - a. Flight planning
 - b. Weather charts
 - c. Navigation
 - d. Navigation aids (NAVAIDs)
 - e. High Altitude Redesign (HAR)
 - f. RNAV/Required Navigation Performance (RNP) and Receiver Autonomous Integrity Monitoring (RAIM) prediction
4. Physiological Training
 - a. Respiration
 - b. Hypoxia
 - c. Effects of prolonged oxygen use
 - d. Decompression sickness
 - e. Vision
 - f. Altitude chamber (optional)
5. High-Altitude Systems and Components
 - a. Oxygen and oxygen equipment
 - b. Pressurization systems
 - c. High-altitude components
6. Aerodynamics and Performance Factors
 - a. Acceleration and deceleration
 - b. Gravity (G)-forces
 - c. Mach Tuck and Mach Critical (turbojet airplanes)
 - d. Swept-wing concept
7. Emergencies
 - a. Decompression
 - b. Donning of oxygen masks
 - c. Failure of oxygen mask or complete loss of oxygen supply/system
 - d. In-flight fire
 - e. Flight into severe turbulence or thunderstorms
 - f. Compressor stalls

Flight Training

1. Preflight Briefing
2. Preflight Planning
 - a. Weather briefing and considerations
 - b. Course plotting
 - c. Airplane Flight Manual (AFM)
 - d. Flight plan
3. Preflight Inspection
 - a. Functional test of oxygen system, including the verification of supply and pressure, regulator operation, oxygen flow, mask fit, and pilot and air traffic control (ATC) communication using mask microphones
4. Engine Start Procedures, Run-up, Takeoff, and Initial Climb
5. Climb to High Altitude and Normal Cruise Operations While Operating Above 25,000 Feet Mean Sea Level (MSL)
6. Emergencies
 - a. Simulated rapid decompression, including the immediate donning of oxygen masks
 - b. Emergency descent
7. Planned Descents
8. Shutdown Procedures
9. Postflight Discussion

Chapter Summary

Transitioning from a non-turbopropeller airplane to a turbopropeller-powered airplane is discussed in this chapter. The major differences are introduced specifically handling, powerplant, and the associated systems. Turbopropeller electrical systems and operational considerations are explained to include starting procedures and high temperature considerations. Training considerations are also discussed and a sample training syllabus is given to show the topics that a pilot should become proficient in when transitioning to a turbopropeller-powered airplane.

Airplane Flying Handbook (FAA-H-8083-3C)

Chapter 16: Transition to Jet-Powered Airplanes

Introduction

This chapter contains an overview of jet-powered airplane operations. The information contained in this chapter provides a useful preparation for, and a supplement to, structured jet airplane qualification training. This chapter provides information on major differences a pilot may encounter when transitioning to jet-powered airplanes. The major differences between jet-powered airplanes and piston-powered airplanes have been addressed in several distinct areas: differences in aerodynamics, systems, and pilot operating procedures. For airplane-specific information, a pilot should refer to the FAA-approved Airplane Flight Manual for that airplane.

Ground Safety

Stepping out on the ramp in the vicinity of jet airplanes requires special caution. There is no propeller to indicate visually whether a jet engine is running. It is easy to inadvertently stray into danger since, even at idle, jet engines are a threat. Enough air is being sucked into the intake to pull a nearby person into the fan. The air coming from the exhaust is hot and moving fast enough to blow a person down.

Pilots operating jet-powered airplanes should exercise caution during taxi and when adding power to start moving. Adding too much power can pull damaging debris up off the ground or cause damage well behind the aircraft. Jet blast when taxiing into parking areas may affect any loose ground equipment.

Jet Engine Basics

A jet engine is a gas turbine with basic cycle of operation; that is, induction, compression, combustion, expansion, and exhaust. Air passes through the intake and enters the compressor section, which is made up of a series of fan blades or “stages.” The first stage, visible from the front of the engine, is the largest diameter and has the biggest blades. Each subsequent stage contains smaller diameter and thinner blades of increasing pitch. The compression in each stage raises the air temperature and pressure. The high-pressure hot air enters the combustion chamber where fuel is added. During engine start, igniters set the fuel air mixture on fire, after which the fire is self-sustaining. The rapidly expanding air flows to the turbine section, which like the compressor section, consists of a series of fan blade stages. The turbine section extracts a portion of the available energy from the airflow to turn a shaft, which drives the compressor. The remaining energy causes rapid air expansion in the nozzle of the tail pipe, accelerates the gas to a high velocity, and produces thrust. [Figure 16-1]

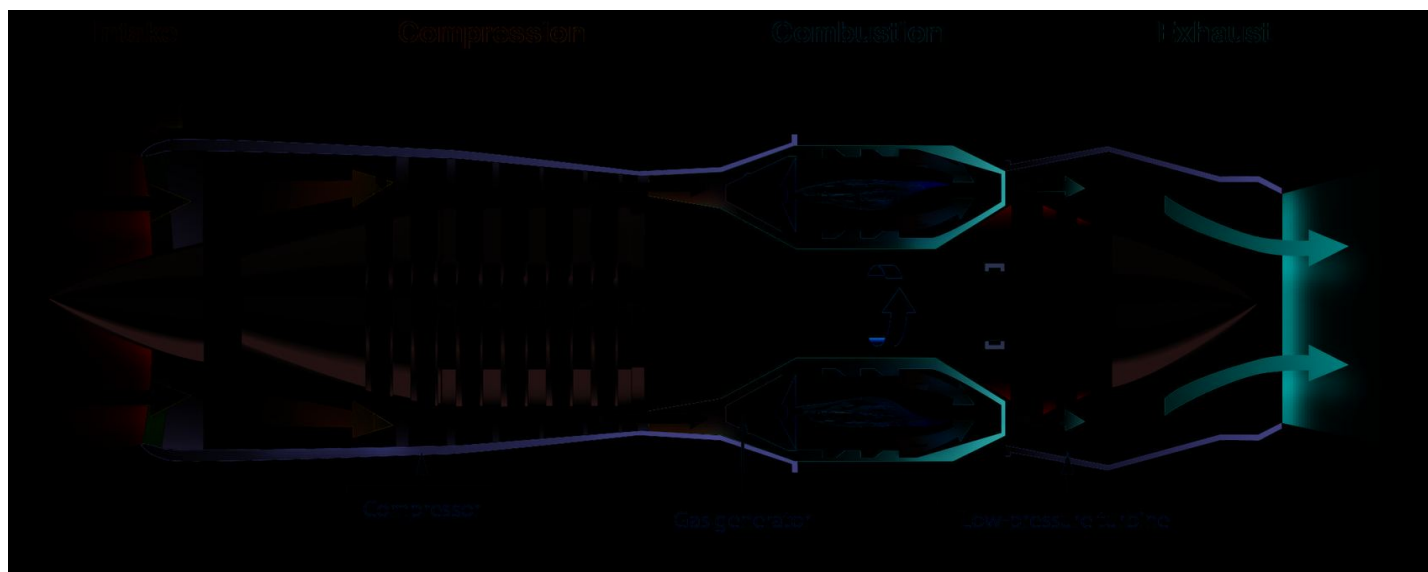


Figure 16-1. Basic turbojet engine.

The large first stage design of a turbofan engine, a ducted fan, diverts some of the air around the engine core. This cooler bypass air produces some of the thrust. The amount of air that bypasses the core compared to the amount compressed for combustion determines a turbofan's bypass ratio. In a turbofan engine, the compressor and turbine sections divide into sub-sections. Each sub-section in the turbine section connects to a specific sub-section of the compressor section via a split-spool shaft. [Figure 16-2]